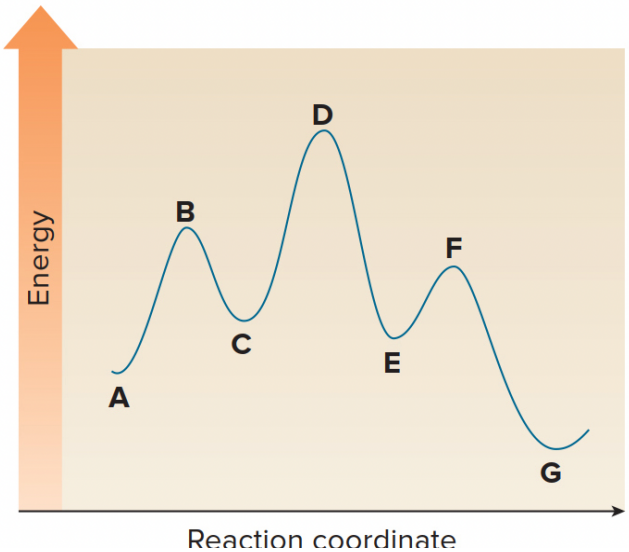
Organic Cume, September 6th, 2024

1. Indicate which factors affect the rate of a reaction.
1) ΔG^0 2) ΔH^0 3) E_a 4) temperature 5) concentration 6) K_{eq} 7) k 8) catalysts

2. 1) Which value corresponds to a negative value of ΔG^0: $K_{eq} = 10^{-2}$ or $K_{eq} = 10^2$?
2) In a unimolecular reaction with five times as much starting material as product at equilibrium, what is the value of K_{eq}? Is ΔG^0 positive or negative?
3) Which value corresponds to a larger K_{eq}: $\Delta G^0 = -8$ kJ/mol or $\Delta G^0 = 20$ kJ/mol?

3. Consider the following energy diagram for the conversion of A to G.
1) Which points on the graph correspond to transition states?
2) Which points on the graph correspond to reactive intermediates?
3) How many steps are present in the reaction mechanism?
4) Label each step of the mechanism as endothermic or exothermic.
5) Label the overall reaction as endothermic or exothermic.

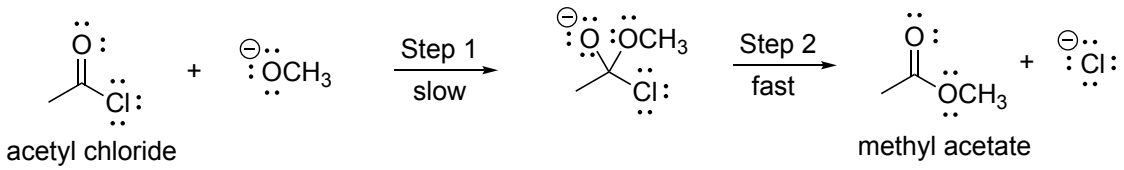


4. Use the attached bond dissociation energy table to calculate ΔH^0 for the following two reactions.

1) $HO\bullet + CH_4 \longrightarrow H_2O + \bullet CH_3$

2) $HBr + CH_3OH \longrightarrow H_2O + CH_3Br$

5. The conversion of acetyl chloride to methyl acetate occurs via the following two-step mechanism:

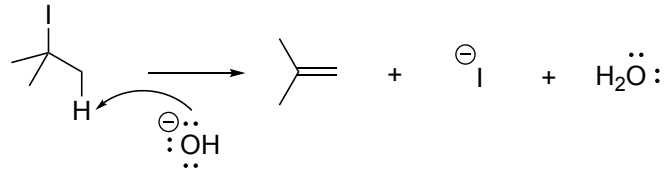


1) Add curved arrows to show the movement of the electrons in each step.
2) Write the rate equation for this reaction, assuming the first step is rate-determining.

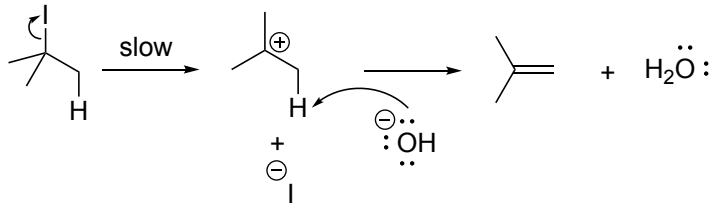
3) If the concentrations of both CH_3COCl and $^{-}OCH_3$ were increased 10 times, what would happen to the rate of the reaction?

6. The conversion of $(CH_3)_3CI$ to $(CH_3)_2C{=}CH_2$ can occur by either a one-step or two-step mechanism, as shown in equations 1 and 2.

1)



2)



a) What rate equation would be observed for the mechanism in equation 1?
b) What rate equation would be observed for the mechanism in equation 2?
c) How can these rate equations be used to show which mechanism is the right one for this reaction?
d) Assuming that equation 2 represents an endothermic reaction and that the product of the rate-determining step is higher in energy than the reactants or products. Draw an energy diagram for this two-step reaction. Label the axes, reactants, and products for each step, and the E_a and ΔH^0 for each step. Label $\Delta H^0_{overall}$.

Table 6.2 Bond Dissociation Energies for Some Common Bonds [A—B → A• + •B]

Bond	$\Delta H°$ kJ/mol	(kcal/mol)
H—Z bonds		
H—F	569	(136)
H—Cl	431	(103)
H—Br	368	(88)
H—I	297	(71)
H—OH	498	(119)
Z—Z bonds		
H—H	435	(104)
F—F	159	(38)
Cl—Cl	242	(58)
Br—Br	192	(46)
I—I	151	(36)
HO—OH	213	(51)
R—H bonds		
CH_3—H	435	(104)
CH_3CH_2—H	410	(98)
$CH_3CH_2CH_2$—H	410	(98)
$(CH_3)_2CH$—H	397	(95)
$(CH_3)_3C$—H	381	(91)
$CH_2{=}CH$—H	435	(104)
HC≡C—H	523	(125)
$CH_2{=}CHCH_2$—H	364	(87)
C_6H_5—H	460	(110)
$C_6H_5CH_2$—H	356	(85)
R—R bonds		
CH_3—CH_3	368	(88)
CH_3—CH_2CH_3	356	(85)
CH_3—$CH{=}CH_2$	385	(92)
CH_3—C≡CH	489	(117)
R—X bonds		
CH_3—F	456	(109)
CH_3—Cl	351	(84)
CH_3—Br	293	(70)
CH_3—I	234	(56)
CH_3CH_2—F	448	(107)
CH_3CH_2—Cl	339	(81)
CH_3CH_2—Br	285	(68)
CH_3CH_2—I	222	(53)
$(CH_3)_2CH$—F	444	(106)
$(CH_3)_2CH$—Cl	335	(80)
$(CH_3)_2CH$—Br	285	(68)
$(CH_3)_2CH$—I	222	(53)
$(CH_3)_3C$—F	444	(106)
$(CH_3)_3C$—Cl	331	(79)
$(CH_3)_3C$—Br	272	(65)
$(CH_3)_3C$—I	209	(50)
R—OH bonds		
CH_3—OH	389	(93)
CH_3CH_2—OH	393	(94)
$CH_3CH_2CH_2$—OH	385	(92)
$(CH_3)_2CH$—OH	401	(96)
$(CH_3)_3C$—OH	401	(96)